

Human-Centered Design Techniques Portfolio

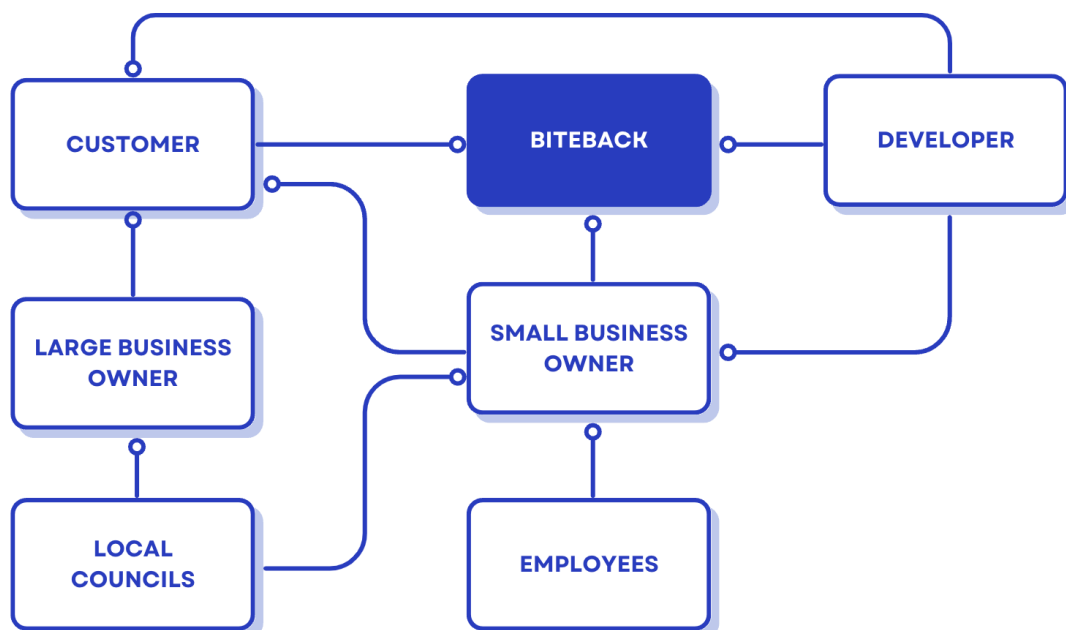
People

Type	Customer	Small Business Owner
Description	The Customer is a member of the general public who wants to find new places to eat, and wants to avoid repeatedly going to larger businesses and franchises.	The Small Business Owner is someone who owns a “local” food business and is struggling to compete with larger businesses in their area.
Needs	They want to find good deals to save themselves money and they want to be able to do this while they are out and on the go.	They want a way to put themselves on the map and attract new customers, while also increasing customer retention by maintaining a close relation with the people who buy from their business.
Example	Michael - Michael is a university student who likes to eat out frequently. He's interested in finding new places to eat and likes to use restaurant loyalty apps to see what deals are available, but he's tired of going to the same spots when he's out and wishes there was an easier way to find less visible food businesses while he's out.	Kelvin - Kelvin just opened his own bubble tea shop in central London. He aspires to introduce the local community to high quality bubble teas. Kelvin has clearly expressed that he wants a way to put himself on the map and get more customers.

Stakeholders

We obviously have our target audience that are directly affected however, there are also other stakeholders that we should consider for this project; This is going to focus on the other stakeholders apart from the target audience. There is obviously the developer team(which is us) that have a stake to deliver a product that is what the customers and small businesses actually want. We also have further stakeholders within the small businesses; there will be employees of the business(that may or may not be the owner) that will need to use this product very frequently; they would like the product to be fast and easy to use. We also have wider stakeholders; there will be larger businesses which would have many locations. They are competitive with smaller businesses and our solution should lead to there being an increased number of customers going to smaller businesses; doing this would be implicitly taking away customers from larger businesses. There are also local councils that would want to have a larger local economy that would want to support efforts like our solution that would strengthen the local economy and help small businesses.

Stakeholder Mapping Diagram



Current state of the users

So when we were examining the current state of our users we made use of design techniques. So first of all we wanted to create user journeys of the problems that they would face during a regular day; these are shown below. After we created and analysed our user journeys we wanted to try to generate ideas using the “How might we...” structure. “How

might we create a way for Micheal to find reliable and up to date deals quickly?”. “How might we allow Kelvin’s deals to reach an audience that would want to take advantage of them?”

The following graphics illustrate Michael and Kelvin’s current user journeys. Take note in particular the difficulties they are facing. Michael is struggling to find a good deal, and needs to settle with either subpar pricing or food. Kelvin, on the other hand, has loads of extra high quality teas at the end of the day that have to go to waste due to the fact that people are uninformed of the deals that are being published.



Initial key insights:

Initial qualitative research through interviews revealed that a majority of small business owners struggled to establish a strong online presence, finding it challenging to compete with larger franchises such as McDonalds, Burger King and KFC. Additionally, several of our

stakeholders expressed interest in implementing a loyalty scheme for customer retention and promoting their business. On the customer side, we interviewed regular users of food-based services and they highlighted, although they are attracted by good deals and loyalties, the tediousness in keeping track of multiple apps/loyalty schemes at the same time posed a significant hurdle. Many also raised concerns about the overwhelming number of choices available, and would often lead them to default to familiar restaurants for the sake of convenience - even when wanting to try something new.

Opportunity statement:

From our stakeholders interviews we also did some further research surrounding the topic of businesses creating apps for user engagement and exposure as well as how users wanted to discover new businesses. Online research yielded results, such as “Cost to build a simple loyalty app is up to £30,000” ([smartworking.io](https://www.smartworking.io)) and also found from our own further surveying that customers generally rated it hard to discover new local food places. We know that larger businesses are able to create their own apps as they have the funding to hire employees to work specifically on the application. However, for smaller businesses we found that they didn't have the expertise to set something up on their own and they didn't have the money to hire people to work on an app specifically. We had gathered from our user interviews that there might be an opportunity for us to connect customers with smaller business owners in a new way that is more cost effective for businesses and more convenient for users along with the data prior that we gathered from online sources. This then led us to our opportunity statement:

“How can we create a cheaper alternative to building a personalised app for businesses to aid in their marketing, whilst maximising their exposure to eager customers willing to try out new places?”

Testing and Validation

Our Method

For the development of our application, we wanted to maintain good lines of communication between us, the developers, and the target audience, i.e. the customers and small business owners. We scheduled regular meetings with them during each milestone, showing them our app, asking them to try new features and listening to their feedback. These gave us a way to structure our development and ensure that we were working towards the goal of providing a solution to our opportunity statement. Here we've listed some of the key insights that we gained from using this iterative development style.

The Analytics Page

Concept

When we pitched the idea of the app to small business owners, we had feedback suggesting ways to analyse and visually see what kind of effects deals and loyalty schemes had on customer behaviour, for example speaking to Kelvin, the bubble tea shop owner, he said: “I

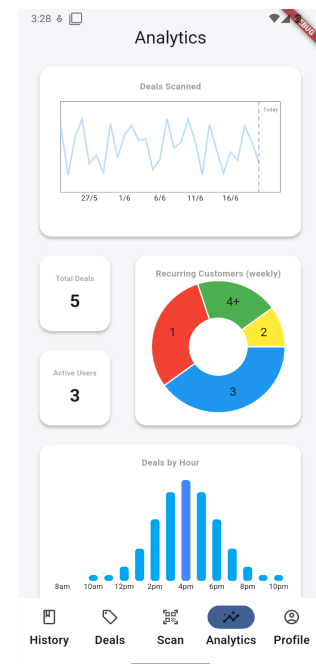
want to see when people are coming in and getting the deals the most. Like when are the people scanning their loyalty cards”.

Feature

With this insight we went away and created the feature and scheduled another meetup with the small business owners to get their opinion.

Experience

After speaking to them a second time, we gained more insight on the types of analytics they'd like to see on the page, for example: “I'd like another graph to display the distribution of how many times customers are returning ... this provides further analytics to see how loyal different customers are” so we created a pie chart to count recurring customers and a timeline of deals scanned to see how much activity they were getting over the course of the last month



Expiring Deals

Concept

After we implemented the deals page for the business owners we got some very interesting feedback from the business owners, who wanted a way to put an expiry date on the deals they create. There were multiple reasons for why:

“I'd like to be able to choose how long different deals last... we advertise some deals that last only on weekends and other deals like stamps are long term.” - Kelvin

“Sometimes you put a deal and then you forget that you put that deal and then you don't have the item left or whatever. Like people are still trying to buy it.” - Ahmed

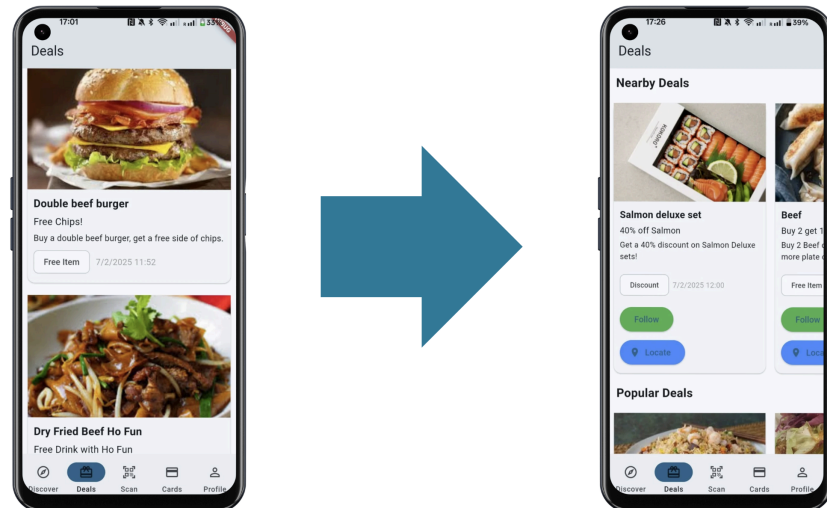
But we also thought that this would really help us with providing the best possible solution to our opportunity statement. We wanted to increase business visibility, and we thought that time based deals were a good way to do so.

Feature

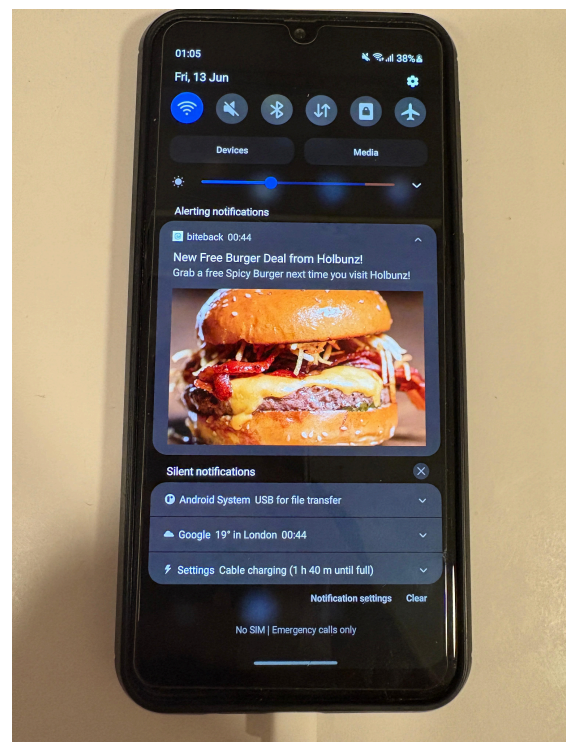
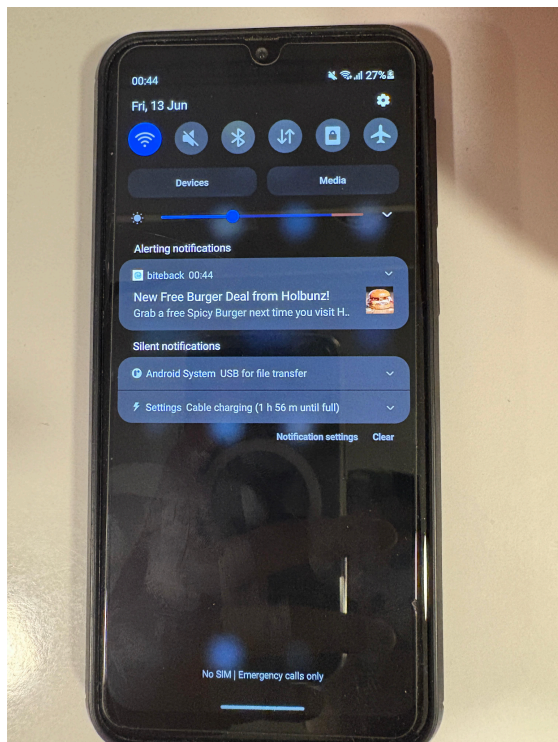
The idea was this: the business owner wants to put up a deal for a particular item. This could be because they want to increase awareness of a new item, try and sell their current stock of the item (if they were in excess) or just as a way to provide good service. When they create the deal they add an expiry time, and this expiry time not only appears on the deal in the customers' deal page but also as a push notification. We can use this expiry time and the user's location to show deals that would be most relevant to them which would then increase the likelihood of them visiting the shop.

Experience

We implemented the features as shown in the screenshots and asked for feedback. The general response was positive. The customers stated that a cleaner UI would be better, in particular they felt that scrolling the deals in a single direction made it more tedious to find and requested a sideways tile scrolling experience as shown below:



We also added the push notification feature as mentioned in our concept:



Nearby Businesses

Concept

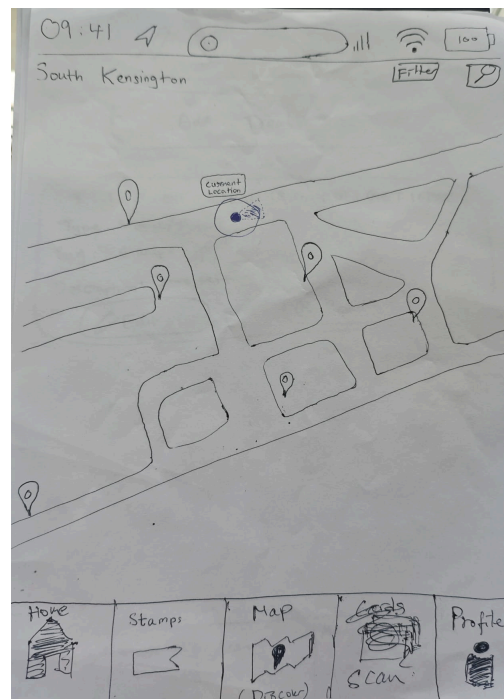
As part of our opportunity statement, we wanted to increase the exposure of local businesses, and so we went out to interview people at the Farmer's Market to get some insight into how people find places to eat.

Feature

The overwhelming feedback was that Google Maps was the favourite way to find nearby places to eat.

"I want to be able to see nearby businesses so that when I'm out I can quickly find a convenient place to eat."

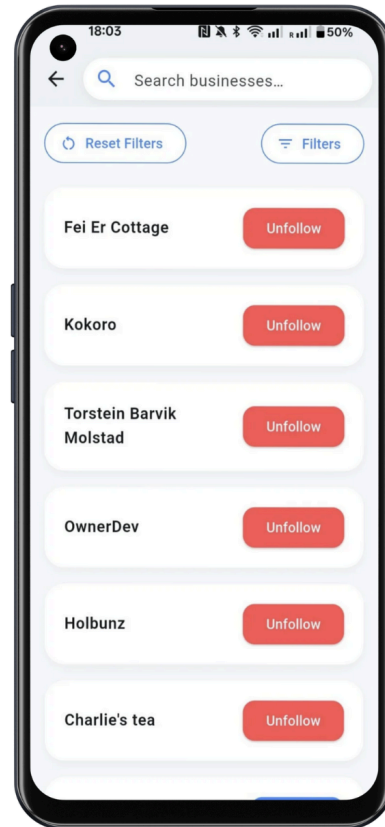
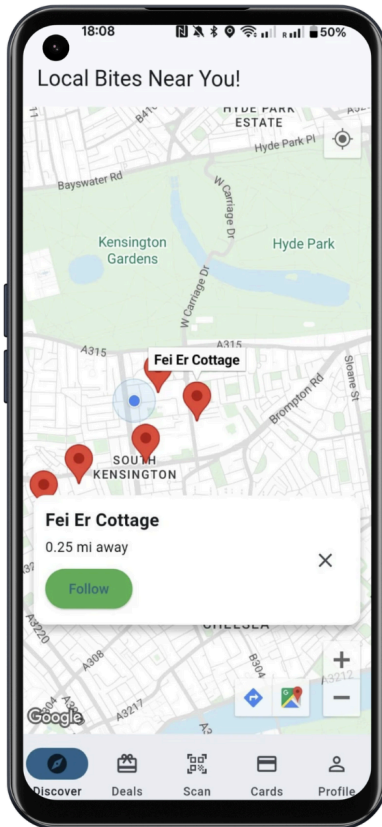
So we created a mockup on paper and showed it to Michael and a few others to get some concrete ideas:



"I want to be able to see nearby businesses so that when I'm out I can quickly find a convenient place to eat." - Michael

Experience

With all of this information we implemented the maps feature:

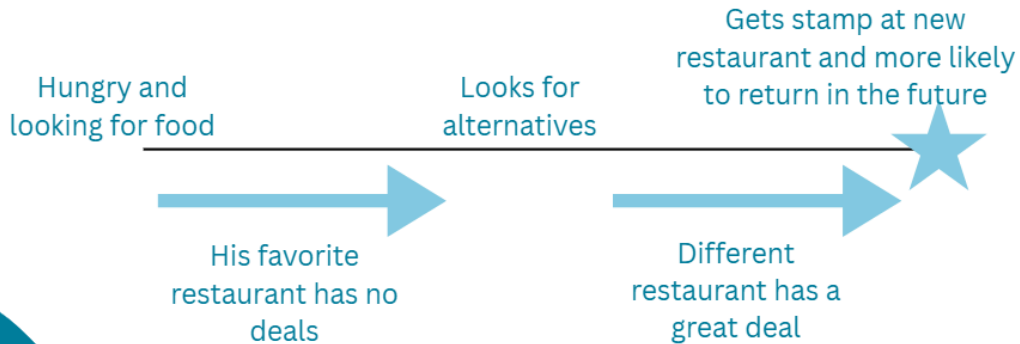


Understanding Impact

Future state of the users

With our product, we see that Michael uses our app to easily find a new, local restaurant to add to his favorites list, becoming a regular customer there. We also see that Kelvin is now able to fully utilise his excess stock by posting deals and through this also gain new customers that are likely to return.

MICHAEL'S NEW JOURNEY



KELVIN'S NEW JOURNEY



Wider impacts:

Our app generates a significant positive impact on the local economy, community as well as the digital and technical innovation landscape. Through encouraging support and discovery of local food businesses, it boosts local spending and keeps cashflow within the community rather than funneling it to large franchises. Furthermore, it empowers local businesses through access to a digital infrastructure, such as loyalty schemes, deals and analytics, while at the same time being lower in cost and expertise - major barriers local business owners

face. Additionally, on the customer side, the app simplifies discovery by promoting personalized deals and reducing decision fatigue. This helps users explore hidden gems while rewarding them for repeat visits. The resulting ecosystem has a thriving local community, where local food vendors are digitally empowered, and a consumer base that can enjoy a richer and more rewarding food experience.